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13.2.4 Chemical properties:

Alkanes are quite unreactive towards most reagents. However under favorable conditions, alkanes undergo the following type of reaction.

Paraffin is the older name for the alkane group family of compounds. This name comes from the Latin which means 'little activity'

1) Combustion:

A combustion reaction is a chemical reaction between a substances and oxygen with evolution of heat and light (usually as a flame). In the presence of sufficient oxygen, alkanes undergoes combustion when ignited and produces carbondioxide and water.

The combustion reaction is expressed as follows

for example:

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O \quad \Delta H° = -890.4 kJ$$

When alkanes burn in insufficient supply of oxygen, they form carbonmonoxide and carbon black.

$$CH_4 + 3O_2 \xrightarrow[1273K]{Ni} 2CO + 4H_2O$$

$$CH_4 + O_2 \longrightarrow C + 2H_2O$$

Evaluate Yourself

6) Write down the combustion reaction of propane whose $\Delta H° = -2220$ kJ

2) Halogenation:

Ahalogenation reaction is the chemical reaction between an alkane and halogen in which one or more hydrogen atoms are substituted by the halogens.

Chlorination and Bromination are two widely used halogenation reactions. Fluorination is too quick and iodination is too slow. Methane reacts with chlorine in the presence of light or when heated as follows.

$$CH_4 + Cl_2 \xrightarrow[Heat]{Light\ or} \underset{\text{methyl chloride}}{CH_3Cl} + HCl$$

$$CH_3Cl + Cl_2 \xrightarrow[Heat]{Light\ or} \underset{\text{methylene chloride}}{CH_3Cl_2} + HCl$$

$$CH_2Cl_2 + Cl_2 \xrightarrow[Heat]{Light\ or} \underset{\text{chloroform}}{CHCl_3} + HCl$$

$$CHCl_3 + Cl_2 \xrightarrow[Heat]{Light\ or} \underset{\text{carbontetrachloride}}{CCl_4} + HCl$$

Mechanism:

The reaction proceeds through the free radical chain mechanism. This mechanism is characterized by three steps initiation, propagation and termination.

i) CHAIN INITITATION: The chain is initiated by UV light leading to homolytic fission of chlorine molecules into free radicals (chlorine atoms).

$$Cl-Cl \longrightarrow 2Cl^{\cdot}$$

Here we choose Cl-Cl bond for fission because C-C & C-H bonds are stronger than Cl-Cl.

ii) **PROPAGATION:** It proceeds as follows,

(a) Chlorine free radial attacks the methane molecule and breaks the C-H bond resulting in the generation of methyl free radical